

Measuring Changes in Pol II Speed & Splicing Time from Total RNA-seq

Jakub Koubele

Progress Report

9th September, 2025

Introduction

- A pre-print of this work is available on BioRxiv.



THE PREPRINT SERVER FOR BIOLOGY

Estimating changes in RNA Polymerase II elongation and intron splicing speeds from total RNA-seq data

 Jakub Koubele,  Andreas Beyer

doi: <https://doi.org/10.1101/2025.08.24.672013>

- A pre-print of this work is available on BioRxiv.



THE PREPRINT SERVER FOR BIOLOGY

Estimating changes in RNA Polymerase II elongation and intron splicing speeds from total RNA-seq data

 Jakub Koubele,  Andreas Beyer

doi: <https://doi.org/10.1101/2025.08.24.672013>

- Presentation outline:
 - Model derivation
 - Fitting and statistical inference
 - Implementation details

Model overview

Model overview

- The model works with total (ribo-minus) RNA-seq data.

Model overview

- The model works with total (ribo-minus) RNA-seq data.
- The model consists of inputs, outputs, and parameters.

Model overview

- The model works with total (ribo-minus) RNA-seq data.
- The model consists of inputs, outputs, and parameters.
- **Inputs** are the vectors of explanatory variables (sample annotation), denoted $\mathbf{x}$.

Model overview

- The model works with total (ribo-minus) RNA-seq data.
- The model consists of inputs, outputs, and parameters.
- **Inputs** are the vectors of explanatory variables (sample annotation), denoted $\mathbf{x}$.
- **Parameters** are the LFC of gene expression, Pol II speed, and intron splicing time.
 - Several intercept terms are also included.

Model overview

- The model works with total (ribo-minus) RNA-seq data.
- The model consists of inputs, outputs, and parameters.
- **Inputs** are the vectors of explanatory variables (sample annotation), denoted $\mathbf{x}$.
- **Parameters** are the LFC of gene expression, Pol II speed, and intron splicing time.
 - Several intercept terms are also included.
- **Outputs** are the predicted exon and intron read counts, and location of reads within introns (coverage).

Total RNA-seq illustration

Total RNA-seq illustration

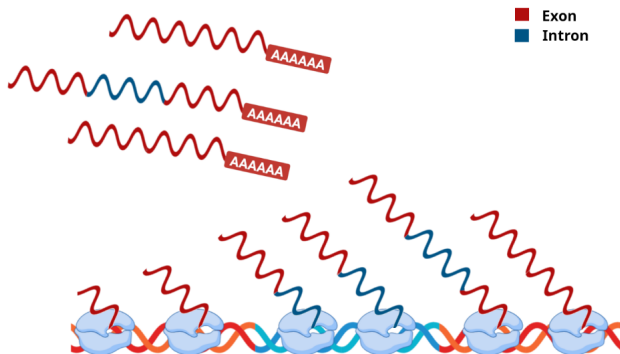


Figure 1: Intronic RNA (in blue) in total RNA-seq may come from two distinct sources: either the intron is currently being transcribed by RNA polymerase, or it was already finished, but not spliced and degraded yet. The finished introns may still be part of the nascent RNA or may come from retained introns in mature RNA.

Data pre-processing

- The reads are aligned to the genome. We construct count matrix for introns (by counting overlapping reads).

- The reads are aligned to the genome. We construct count matrix for introns (by counting overlapping reads).
 - For each intron, we also record its coverage (i.e. read histogram).

- The reads are aligned to the genome. We construct count matrix for introns (by counting overlapping reads).
 - For each intron, we also record its coverage (i.e. read histogram).
- Reads not overlapping with any intron are aligned to the transcriptome, and used to construct a gene-level exon count matrix.

Modeled gene properties

Modeled gene properties

- **Expression rate** (gene-specific):

$$\eta = \eta_0 \exp(\mathbf{x}^\top \boldsymbol{\alpha})$$

The unit is $[\eta] = \text{polymerases} \cdot \text{s}^{-1}$. The parameter $\boldsymbol{\alpha}$ is a vector with the LFC of expression rate.

Modeled gene properties

- **Expression rate** (gene-specific):

$$\eta = \eta_0 \exp(\mathbf{x}^\top \boldsymbol{\alpha})$$

The unit is $[\eta] = \text{polymerases} \cdot \text{s}^{-1}$. The parameter $\boldsymbol{\alpha}$ is a vector with the LFC of expression rate.

- **Pol II speed** (intron-specific):

$$v = v_0 \exp(\mathbf{x}^\top \boldsymbol{\beta})$$

The unit is $[v] = \text{bp} \cdot \text{s}^{-1}$. The parameter $\boldsymbol{\beta}$ is a vector with the LFC of Pol II speed.

Modeled gene properties

- **Expression rate** (gene-specific):

$$\eta = \eta_0 \exp(\mathbf{x}^\top \boldsymbol{\alpha})$$

The unit is $[\eta] = \text{polymerases} \cdot \text{s}^{-1}$. The parameter $\boldsymbol{\alpha}$ is a vector with the LFC of expression rate.

- **Pol II speed** (intron-specific):

$$v = v_0 \exp(\mathbf{x}^\top \boldsymbol{\beta})$$

The unit is $[v] = \text{bp} \cdot \text{s}^{-1}$. The parameter $\boldsymbol{\beta}$ is a vector with the LFC of Pol II speed.

- **Splicing time** (intron-specific):

$$\tau = \tau_0 \exp(\mathbf{x}^\top \boldsymbol{\gamma})$$

The unit is $[\tau] = \text{s}$. The parameter $\boldsymbol{\gamma}$ is a vector with the LFC of splicing time.

Predicting exon read count

Predicting exon read count

- We predict the exon read count as

$$\mu = \exp \left(\text{intercept}_{\text{exons}} + \mathbf{x}^T \boldsymbol{\alpha} \right)$$

i.e. the exon reads simply scale with the gene expression rate.

Predicting exon read count

- We predict the exon read count as

$$\mu = \exp \left(\text{intercept}_{\text{exons}} + \mathbf{x}^T \boldsymbol{\alpha} \right)$$

i.e. the exon reads simply scale with the gene expression rate.

- Two caveats:
 - Small fraction of exonic reads are from nascent RNA; their amount thus also depend on Pol II speed.

Predicting exon read count

- We predict the exon read count as

$$\mu = \exp \left(\text{intercept}_{\text{exons}} + \mathbf{x}^T \boldsymbol{\alpha} \right)$$

i.e. the exon reads simply scale with the gene expression rate.

- Two caveats:
 - Small fraction of exonic reads are from nascent RNA; their amount thus also depend on Pol II speed.
 - Transcript abundance is also affected by the degradation rate (transcript half-life).

Transcribing and unspliced introns

Transcribing and unspliced introns

- The intronic reads come from two sources: currently transcribing introns, and finished introns which are not spliced yet.

Transcribing and unspliced introns

- The intronic reads come from two sources: currently transcribing introns, and finished introns which are not spliced yet.
- Number of reads from transcribing introns scales with the gene expression rate, and with the time needed to transcribe the intron:

$$n_{\text{transcribing}} \propto \eta L v^{-1}$$

where L the intron length, $[L] = \text{bp}$.

Transcribing and unspliced introns

- The intronic reads come from two sources: currently transcribing introns, and finished introns which are not spliced yet.
- Number of reads from transcribing introns scales with the gene expression rate, and with the time needed to transcribe the intron:

$$n_{\text{transcribing}} \propto \eta L v^{-1}$$

where L the intron length, $[L] = \text{bp}$.

- Number of reads from unspliced introns scales with the gene expression rate, and with the time needed to splice out the intron:

$$n_{\text{unspliced}} \propto \eta \tau$$

Predicting intron read count

Predicting intron read count

- The number of intronic reads depends on transcribing and unspliced introns as

$$\nu \propto n_{\text{transcribing}} + 2 \cdot n_{\text{unspliced}}$$

Predicting intron read count

- The number of intronic reads depends on transcribing and unspliced introns as

$$\nu \propto n_{\text{transcribing}} + 2 \cdot n_{\text{unspliced}}$$

- After plugging in the values from gene properties definitions, we get:

Predicting intron read count

- The number of intronic reads depends on transcribing and unspliced introns as

$$\nu \propto n_{\text{transcribing}} + 2 \cdot n_{\text{unspliced}}$$

- After plugging in the values from gene properties definitions, we get:

$$\begin{aligned} \nu \propto & \exp \left(\log(2\tau_0\eta_0) + \theta + \mathbf{x}^T (\boldsymbol{\alpha} - \boldsymbol{\beta}) \right) \\ & + \exp \left(\log(2\tau_0\eta_0) + \mathbf{x}^T (\boldsymbol{\alpha} + \boldsymbol{\gamma}) \right) \end{aligned}$$

where

$$\theta = \log \left(\frac{1}{2} L v_0^{-1} \tau_0^{-1} \right)$$

Predicting intron read count

- The number of intronic reads depends on transcribing and unspliced introns as

$$\nu \propto n_{\text{transcribing}} + 2 \cdot n_{\text{unspliced}}$$

- After plugging in the values from gene properties definitions, we get:

$$\begin{aligned} \nu \propto & \exp \left(\log(2\tau_0\eta_0) + \theta + \mathbf{x}^T(\boldsymbol{\alpha} - \boldsymbol{\beta}) \right) \\ & + \exp \left(\log(2\tau_0\eta_0) + \mathbf{x}^T(\boldsymbol{\alpha} + \boldsymbol{\gamma}) \right) \end{aligned}$$

where

$$\theta = \log \left(\frac{1}{2} L \nu_0^{-1} \tau_0^{-1} \right)$$

- By introducing an intercept term, we obtain:

$$\nu = \exp(\text{intercept}_{\text{intron}} + \theta + \mathbf{x}^T(\boldsymbol{\alpha} - \boldsymbol{\beta})) + \exp(\text{intercept}_{\text{intron}} + \mathbf{x}^T(\boldsymbol{\alpha} + \boldsymbol{\gamma}))$$

Effect of Pol II speed change illustration

Effect of Pol II speed change illustration

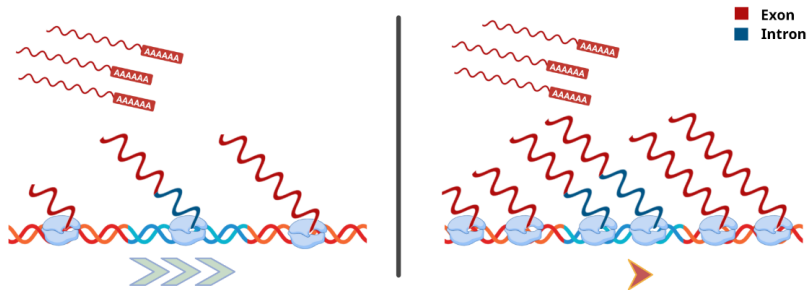


Figure 2: Effect of the change in Pol II speed. In the condition with faster polymerase (left), there is less polymerases currently transcribing a gene, compared to the condition with slower polymerase (right). Note that the amount of mature RNA is not affected by the change in elongation speed.

Intron coverage density

Intron coverage density

- For simplicity, assume the intron to be a continuous interval $[0, 1]$, oriented in the 5' to 3' direction.

Intron coverage density

- For simplicity, assume the intron to be a continuous interval $[0, 1]$, oriented in the 5' to 3' direction.
- Unspliced introns can produce reads anywhere in the intron. Their location then follows a uniform distribution, with the density

$$p_U(r) = 1, \quad r \in [0, 1]$$

Intron coverage density

- For simplicity, assume the intron to be a continuous interval $[0, 1]$, oriented in the 5' to 3' direction.
- Unspliced introns can produce reads anywhere in the intron. Their location then follows a uniform distribution, with the density

$$p_U(r) = 1, \quad r \in [0, 1]$$

- Transcribing introns can produce reads only in the part of the intron that was already transcribed. We model their location by a decreasing triangular density

$$p_T(r) = 2 - 2r, \quad r \in [0, 1]$$

Intron coverage density

- For simplicity, assume the intron to be a continuous interval $[0, 1]$, oriented in the 5' to 3' direction.
- Unspliced introns can produce reads anywhere in the intron. Their location then follows a uniform distribution, with the density

$$p_U(r) = 1, \quad r \in [0, 1]$$

- Transcribing introns can produce reads only in the part of the intron that was already transcribed. We model their location by a decreasing triangular density

$$p_T(r) = 2 - 2r, \quad r \in [0, 1]$$

- The overall distribution of intronic reads is then a mixture of the uniform and triangular components, with the density

$$p(r) = \pi p_T(r) + (1 - \pi) p_U(r) = 1 + \pi - 2\pi r, \quad r \in [0, 1]$$

Where $\pi \in [0, 1]$ is the mixture weight of the triangular distribution.

Intron coverage density illustration

Intron coverage density illustration

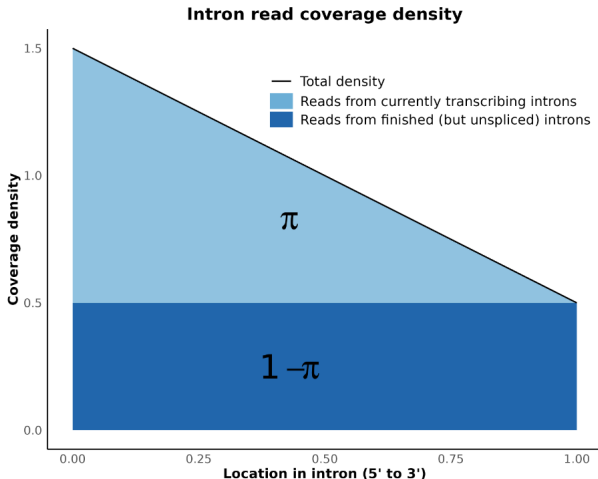


Figure 3: Density function of the intronic read location. The resulting distribution is a mixture of a decreasing triangular distribution, with a weight π , and a uniform distribution, with a weight $(1 - \pi)$.

Predicting intron reads location

Predicting intron reads location

- The mixture parameter π depends on the number of reads from transcribing and unspliced introns:

$$\pi = \frac{n_{\text{transcribing}}}{n_{\text{transcribing}} + 2 \cdot n_{\text{unspliced}}}$$

Predicting intron reads location

- The mixture parameter π depends on the number of reads from transcribing and unspliced introns:

$$\pi = \frac{n_{\text{transcribing}}}{n_{\text{transcribing}} + 2 \cdot n_{\text{unspliced}}}$$

- After plugging in the formulas from definitions, we arrive to

$$\pi = \frac{\exp(\theta - \mathbf{x}^T(\boldsymbol{\beta} + \boldsymbol{\gamma}))}{1 + \exp(\theta - \mathbf{x}^T(\boldsymbol{\beta} + \boldsymbol{\gamma}))} = \sigma(\theta - \mathbf{x}^T(\boldsymbol{\beta} + \boldsymbol{\gamma}))$$

where $\sigma(\cdot)$ denotes the logistic (sigmoid) function.

Fitting the model

Fitting the model

- The model parameters are fitted by maximum likelihood estimation (MLE).

Fitting the model

- The model parameters are fitted by maximum likelihood estimation (MLE).
- Loss function (negative log-likelihood) consists of Poisson losses for count data (intron and exon read counts) and loss from intronic reads locations.

Fitting the model

- The model parameters are fitted by maximum likelihood estimation (MLE).
- Loss function (negative log-likelihood) consists of Poisson losses for count data (intron and exon read counts) and loss from intronic reads locations.
- Optimization is performed by quasi-Newton algorithm (L-BFGS).

Fitting the model

- The model parameters are fitted by maximum likelihood estimation (MLE).
- Loss function (negative log-likelihood) consists of Poisson losses for count data (intron and exon read counts) and loss from intronic reads locations.
- Optimization is performed by quasi-Newton algorithm (L-BFGS).
- Problem: the loss function is not convex (due to the reads location term).

Fitting the model

- The model parameters are fitted by maximum likelihood estimation (MLE).
- Loss function (negative log-likelihood) consists of Poisson losses for count data (intron and exon read counts) and loss from intronic reads locations.
- Optimization is performed by quasi-Newton algorithm (L-BFGS).
- Problem: the loss function is not convex (due to the reads location term).
 - However, a semi-analytical solution exists for a simple case when $\mathbf{x}$ consists of a single factor. In this case, loss minimum is unique.
 - Finding guarantees (or counter-examples) for more complex settings remains an open problem.

Hypothesis testing

- **Likelihood-ratio test (LRT)**: Fit a restricted model (e.g., with some LFC set to 0). Difference of log-likelihood vs. unrestricted model is used in the χ^2 test.

Hypothesis testing

- **Likelihood-ratio test (LRT):** Fit a restricted model (e.g., with some LFC set to 0). Difference of log-likelihood vs. unrestricted model is used in the χ^2 test.
- **Wald test:** Assumes a Normal distribution of the parameters under H_0 . The covariance matrix is estimated as the inverse of the loss Hessian.
 - Less reliable than LRT (the normality assumption is often violated), but faster to compute.

Implementation

Implementation

- Code is available on GitHub:
<https://github.com/jkoubele/pol-ii-speed>

Implementation

- Code is available on GitHub:
<https://github.com/jkoubele/pol-ii-speed>
- Model is implemented in PyTorch.  PyTorch

Implementation

- Code is available on GitHub:
<https://github.com/jkoubele/pol-ii-speed>
- Model is implemented in PyTorch.  PyTorch
- The whole pipeline is orchestrated by Nextflow.



Implementation

- Code is available on GitHub:
<https://github.com/jkoubele/pol-ii-speed>
- Model is implemented in PyTorch.  PyTorch
- The whole pipeline is orchestrated by Nextflow.



- Containerization via Docker (with images hosted on Docker Hub).



The End

Thank you for your attention!



Illustrations were created using BioRender.